#10
JOHN BECK: BUCKMINSTER FULLER AND THE POLITICS OF SHELTER

In Nathaniel Hawthorne's *The House of the Seven Gables* (1851),[1] Holgrave, the free-thinking daguerreotypist lodging with Hepzibah Pyncheon, the reclusive owner of the seventeenth-century house of the title, is pontificating to Phoebe, Hepzibah's young country cousin. 'Shall we never get rid of this past!' he complains, 'It lies upon the Present like a giant's dead body!' The burden of the past is everywhere a tyranny of habit and repetition, housing the present in its dilapidated forms. After a gloomy litany of restrictions imposed by the 'Dead Man's icy hand', however, Holgrave soars in rebellion as he envisions the day 'when no man shall build his house for posterity':

> If each generation were allowed and expected to build its own houses, that single change, comparatively unimportant in itself, would imply almost every reform which society is now suffering for. I doubt whether even our public edifices – our capitols, state-houses, court-house, city-halls, and churches – ought to be built of such permanent materials as stone or brick. It were better that they should crumble to ruin, once in twenty years, or thereabouts, as a hint to the people to examine into and reform the institutions which they symbolize.[2]

Buckminster Fuller (1895–1983) is an inheritor of a powerful tradition of New England nonconformism, characterized most memorably by the Transcendentalists of the mid-nineteenth century, which included Fuller's great aunt Margaret Fuller, and of which Hawthorne was a sceptical satellite. A hybrid of German idealism, Jeffersonian republicanism and Yankee self-help, Transcendentalism was didactic, utopian, organicist, supremely individualistic. The children of revolutionaries, its adherents saw the USA as a democracy and as a nation, as a source of moral and social salvation, philosophical argument bleeding effortlessly, if not always intentionally, into a discourse of expansionist imperialism. Reading the USA's founding principles as natural serves to universalize the US model and de-natures all other ways of life. While this is a familiar argument, the conjunction of the revolutionary democratic impulse and the celebration of a manifest destiny inheres in Fuller's vision of an emancipatory 'comprehensive design science', and might explain why he was courted by both the hippies and the Marines.

'I was born cross-eyed', Fuller writes. It was four years before his acute long-sightedness was corrected with glasses. 'Until four I could see only large patterns, houses, trees, outlines of people with blurred colouring [...]. I did not see a human eye or a

1 Nathaniel Hawthorne, *The House of the Seven Gables*, New York: Norton, 1967 (originally published 1851).
2 *Ibid.*, pp. 182–184.

teardrop or a human hair until I was four."[3] We may deduce two things from this. Firstly, the useful intervention of technology to adjust natural dysfunction, as in Herman Melville's *The Confidence Man* (1857): 'Nature made me blind and would have kept me so. My oculist counterplotted her'. Secondly, that despite Fuller's 'new ability to apprehend details, my childhood's spontaneous dependence only upon big pattern clues has persisted'.[4]

This mix of romantic enchantment counterplotted with technological realignment embeds Fuller's industrial innovations within an organic universe where human ingenuity confirms a sense of belonging and sovereignty in the face of an indifferent environment. It is quite an ingenious manoeuvre which enables industrial manufacture on the largest scale to be presented as coterminous with natural process. Fuller's structures, in following the most basic mathematical models – triangle, tetrahedron, icosahedron – are thereby confirming their similitude to molecular forms – the 'big pattern clues' are also the details.

Fuller served in the Navy between 1917 and 1922, then went into the construction business with his father-in-law until the company's failure five years later. 1927, the year Lindbergh flew the Atlantic, was Fuller's much vaunted 'year of silence', when, so he claimed, he spoke to no one, not even his wife. The mass of drawings, designs and writing produced during this year – much of which was privately printed in an edition of 200 and distributed to friends and possible sponsors in 1928 as *4D* – reveal Fuller's main ideas to be in place. By 1938, with the publication of Fuller's first real book, *Nine Chains to the Moon*,[5] the vision has taken the form of a dwelling-led, full-on critique of corporate capitalism. During this eleven years the USA had enjoyed unprecedented prosperity and endured unparalleled economic hardship. Fuller's designs are a consequence of both. The Depression confirmed for Fuller what his own lack of commercial success had revealed: capitalism worked inefficiently, self-interest and short-term gain inhibiting experimentation and truly beneficial technological change. Housing suffered under the tyranny of the architect, planner, manager, local politician and government bureaucrat. Fuller believed that efficiency of design would reveal these hindrances for what they really were: obstacles placed by backward looking, power hungry beneficiaries of a corrupt system.

Fuller's vision was of a technological utopia based on mass-produced, prefabricated shelter, airlifted to site in canisters and retailing at around the same price as a top of the range family automobile. This idea is first formulated with the 4D towers (4D for 4-

3 Richard Buckminster Fuller, *Utopia or Oblivion: the prospects for humanity*, London: Allen Lane, 1970, p. 11. For other studies of Fuller's work see: J. Baldwin, *BuckyWorks: Buckminster Fuller's ideas for today*, New York: Wiley, 1996; M. Ben-Eli (editor), R. Buckminster Fuller retrospective, special issue of *Architectural Design*, 42, no. 12, 1972; Richard Buckminster Fuller and R. Marks, *The Dymaxion World of Buckminster Fuller*, New York: Anchor, 1960, 1973; H. Kenner, *Bucky: a guided tour of Buckminster Fuller*, New York: Morrow, 1973; Lloyd Steven Sieden, *Buckminster Fuller's Universe: an appreciation*, New York: Plenum, 1989. For more information about the work of Buckminster Fuller contact: The Buckminster Fuller Institute, 2040 Alameda Padre Serra, Suite 224, Santa Barbara, CA 93103, USA (tel. (805) 962-0022).
4 Richard Buckminster Fuller, *Utopia or Oblivion: the prospects for humanity*, *op. cit.*, note 3, p. 11. Herman Melville, *The Confidence Man: his masquerade*, New York: The Library of America, 1984, p. 956 (originally published 1857).
5 Richard Buckminster Fuller, *Nine Chains to the Moon*, London: Cape, 1973 (originally published 1938).

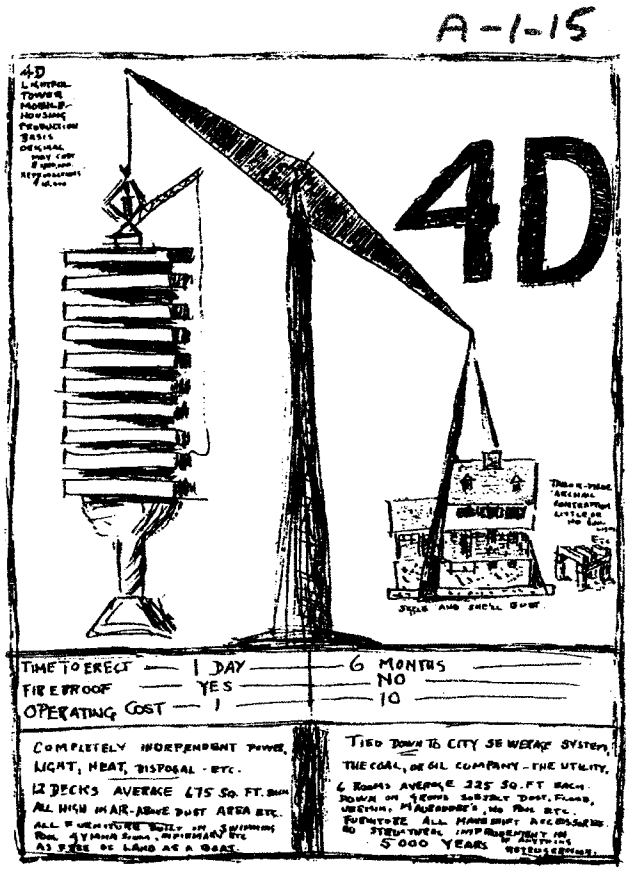


figure 10.1
SKETCH BY FULLER FROM 1927 COMPARING A CONVENTIONAL HOUSE TO HIS TWELVE-DECK TOWER. ©1960, ESTATE OF BUCKMINSTER FULLER

dimensional thinking – an Einsteinian space–time influence) and the Dymaxion House (Dynamic + Maximum + ion) of 1927. A model of this house, mast-supported, and made of high-stretch alloys and plastics developed for the aircraft industry, was first exhibited in a Chicago department store in 1929, and its design published in *Architectural Forum* and *Shelter* in 1932 (Figure 10.1). Within twenty-four hours (the time taken for delivery, Fuller promised), the new occupant could expect the elimination of 'drudgery, selfishness, exploitation, politics and centralized control'.[6]

The scales in Fuller's primitive drawing tell the oppressive truth about conventional housing: heavy, expensive, time-consuming, dirty and vulnerable. By contrast, Fuller's twelve storey tower is 'completely independent', self-regulating, raised above the squalor of ground level, 'as free of land as a boat' (Figure 10.2). Weight is the enemy of freedom because it is resistant to movement, and the immovable demands maintenance and preservation. For Fuller, such dwellings irrationally resist the energies of space and time

6 Martin Pawley, *Buckminster Fuller*, London: Trefoil, 1990, p. 56.

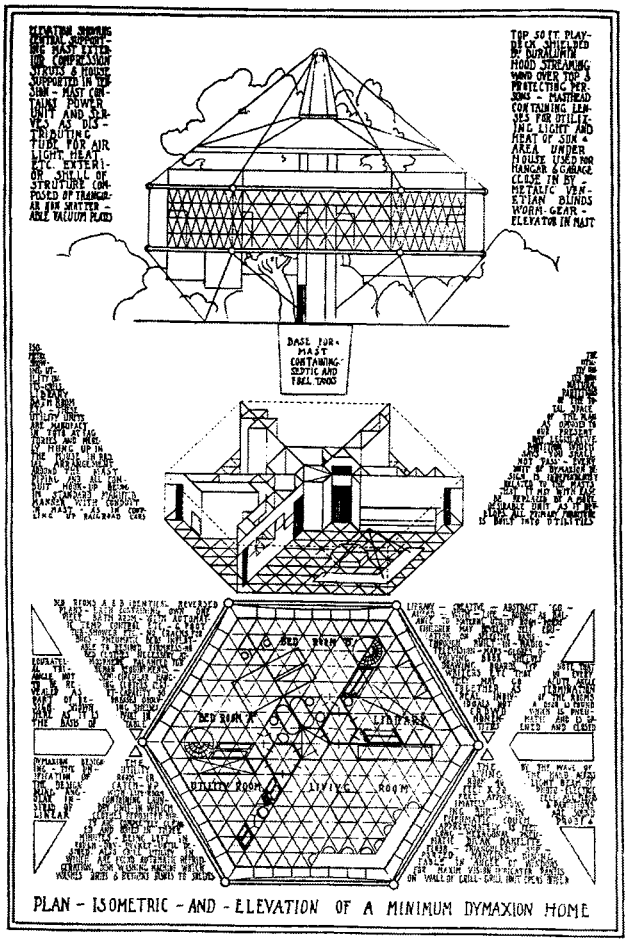


figure 10.2
PLAN, ISOMETRIC, AND ELEVATION OF THE 1927 DYMAXION HOUSE.
©1960, ESTATE OF BUCKMINSTER FULLER

and are no more than monuments to an arrogant and untenable belief in fixity. Fuller is not interested in the least in architecture as a profession or as an art form, he is interested in social action: 'In architecture "form" is a noun; in industry, "form" is a verb. Industry is concerned with DOING, whereas architecture has been engrossed with making replicas of end results of what people have industrially demonstrated in the past'.[7] The modern movement, for Fuller, went no further than to update the shell of an obsolete artefact – the permanent dwelling. As such, he would agree with Roger Scruton's point that the functionalist theories of modern architecture 'have been used not to condemn but to articulate aesthetic values'.[8]

Given the remarkable impact on daily life and social organization effected by the motor car as a result of Henry Ford's production methods, Fuller's hopes for his house are

7 Richard Buckminster Fuller, *op. cit.*, note 5, p. 41.
8 Roger Scruton, *The Aesthetics of Architecture*, London: Methuen, 1979, p. 38.

perhaps not unreasonable. And his admission that it would cost $100 million to produce a prototype is less extravagant considering it cost Ford $43 million to develop the Model-A.[9] Fuller had more than an eye on the car industry when designing his own future for the USA. Nothing confirmed the practical wisdom of the inventor/pioneer like Ford's production line, introduced in 1913. As a consequence, by 1924 six cars could be built for what it cost to make one in 1904. Fuller was not alone in seeing possibilities for dwelling machines. Walter Gropius compared figures in 1938: between 1913 and 1926 the average cost of building a house rose by 200 per cent, the cost of living by 178 per cent. The cost of cars had fallen in the same period by 22 per cent, the cost of Ford cars by 50 per cent.[10] The lesson is clear.

The car always meant more than transportation in the USA. It embodied foundational American values: individual freedom, perpetual movement, self-transformation through geographical displacement. When Emerson said that 'everything good is on the highway', he was declaring an expansive faith that Ford merely made manifest. The liberty suggested by such rootlessness is at the heart of Fuller's houses. His rejection of weight and mass in architecture in favour of tensional structure is a rejection of monumentality and permanence in preference for the contingent and ephemeral. This is a state of mind as well as a design strategy, signalling an American suspicion of fixed and immutable form which can be traced back at least as far as Jefferson.

Fuller's mobile housing is a proposal for social revolution which relies for its validity upon peculiarly American readings of open space as somehow free for occupation, of land as horizonless, of the innate benefits of mobility as an agent of individual and civic well-being. One of the greatest fears harboured by Americans since the time of the Revolution was that the republic would stagnate, become fixed and immobile, degenerating into nothing but another version of European rigidity and corruption. My opening passage from Hawthorne's novel is an articulation of Jefferson's suggestion that a revolution was needed every twenty years in order to keep the country fresh and young, a project of perpetual reinvention. Fuller's frequent and sustained attack on the meaninglessness of modern life, its routine, its creative barrenness, the alienation generated by its cramped and filthy cities, is a Jeffersonian attack. Cantilevered onto Jefferson's pastoralism is a utopian vision of manufactured reconstruction, a form of industrial production that will bring about the end of capitalism through the elimination of the false needs that an economy of consumption endlessly generates. A contempt for place is relieved by a libertarian notion of space. The globe becomes an open prairie and air travel gives access to all areas. The Dymaxion house is a cross between the covered wagon and the tepee, shelter for a wandering people. The nineteenth-century notion of the frontier as a safety valve against urban disaffection and individual suffocation is renovated through modern technology and the American self is enabled to return to his or her proper state – that of the nomad.

9 Martin Pawley, *op. cit.*, note 6, p. 56.
10 Martin Pawley, *Architecture versus Housing*, London: Studio Vista, 1971, p. 50.

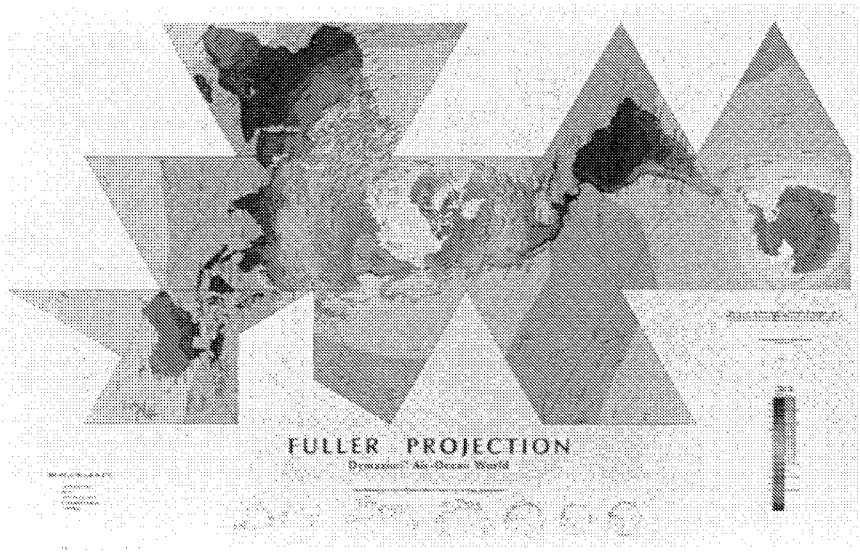
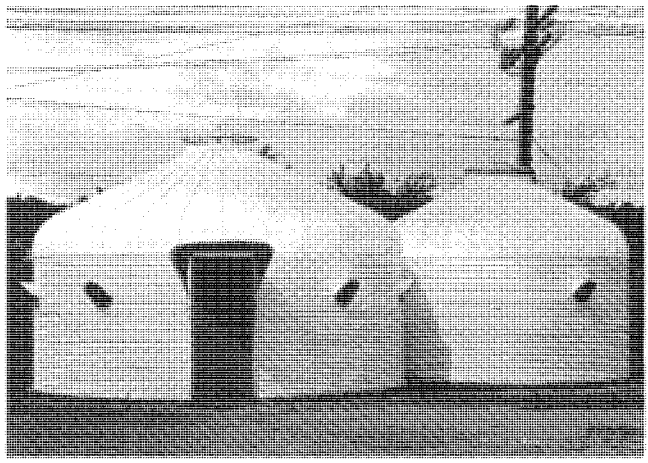


figure 10.3
WORLD MAP ON DYMAXION PROJECTION.
©1960, ESTATE OF BUCKMINSTER FULLER

A nomadic people is free-ranging, unsettled, and therefore open to adaptation as conditions change. Unchained from fixed horizons and habits of thought, physical and intellectual structures are dismantled in favour of exploration and experimentation. A nomadic community, organized around the principles of mobility and necessity, must be flexible and therefore non-hierarchical. Bonds formed through mutual interest can be more easily broken than when society is confined by economic, racial or class imperatives.

Perhaps Fuller's most vivid display of a nomad sensibility is his world map (Figure 10.3). A rejection of Eurocentric deformations of global representation, Fuller's map manages to chart the Earth's surface with minimum distortion. Keeping the shape constant, the continents can be shifted around to reveal different global patterns and networks. In the Air-Ocean version, the land masses of the planet seem peculiarly clustered together. The shortest route between places is across the Arctic. This is the world of air transport, a smooth space of undirected movement. The first map was published around 1940, although preliminary sketches go back at least to 1936. The houses are wedded to a cartography of delimited nomadic space.

While the USA has made a virtue of rootlessness and celebrated geographical wandering as a metonym for national liberty, the American nomad is fraught with contradictions. It is, in fact, the indigenous population of much of the USA which should most properly be considered nomadic, with its attendant fluid and communal notion of land rights. It is this very vagrancy of the Indians which enabled settlers to legally claim land to be free, that is, unsettled and available. Equating settlement with ownership leaves Indian conceptions of habitation literally ungrounded. The white nomad – the pioneer – is, for Americans, the necessary but temporary phase which precedes the staking out of property boundaries. The American nomad is, then, the



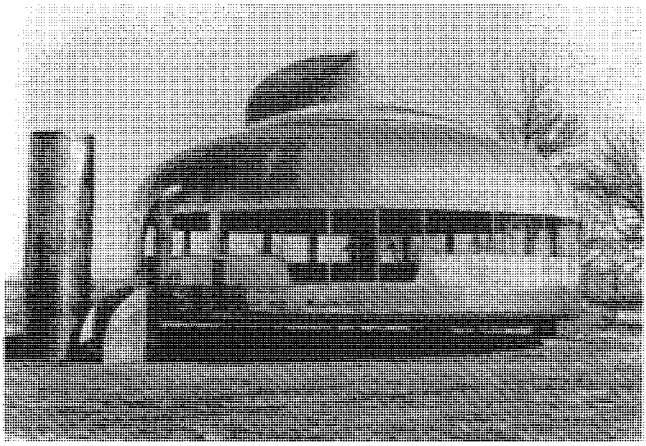
carrier of the very system of restriction he or she symbolically stands against. As Paul Virilio has remarked, 'revolution is movement, but movement is not revolution'.[11]

In rejecting settlement in terms of property ownership, Fuller is rejecting the bourgeois revolution of 1776 and reinstating a prior notion of land as smooth space. On the other hand, this shaking off of boundaries serves to open up all space for colonization and domination. The movements of multinational corporations have confirmed this in their unhindered vagrancy around the planet, opportunistically alighting on favourable sites where labour, land and materials are cheapest. In terms of labour, the idea of mobility is promoted by business but is largely impossible in practice, enabling a loosening of responsibility on the part of companies no longer rooted in local communities. Workers, of course, are not able to follow jobs as easily as the ideology of the free market suggests.

Fuller's reasoning, then, assumes an ideal liberal democracy which his plans in themselves cannot create. The construction of the mobile dwelling industry might, indeed, engender such a society, but how is business to be persuaded that it should invest in an industry which will bring about the end of its hegemony? With Fuller, as with so many liberal progressive theories of social change through planning and technique, there is no adequate vocabulary with which to deal with the issue of power and, in this silence, power moves in and takes over nomadic thinking as its own domain.

Fuller liked to claim that he learned everything he needed to know about design and engineering in the Navy. Those brief years of military service also signal the beginning of a sporadic but lifelong collaboration with the US armed forces. During the Second World War Fuller converted steel grain bins into mobile military dwellings, which he called the Dymaxion Deployment Unit (DDU) (Figure 10.4). Hundreds were used in the Pacific and the Persian Gulf. After the war he turned to a new domestic dwelling design, the Dymaxion Dwelling Machine, or Wichita House as it became known

11 Paul Virilio, *Speed and Politics: an essay on dromology* (trans. Mark Polizzotti), New York: Semiotext(e), 1986, p. 18.



top figure 10.5
THE DYMAXION DWELLING MACHINE, OR WICHITA HOUSE, OF 1947, SHOWING THE CYLINDER WHICH CONTAINED ALL THE COMPONENT PARTS. ©1960, ESTATE OF BUCKMINSTER FULLER

left figure 10.4
THE DYMAXION DEPLOYMENT UNIT (DDU), MANUFACTURED BY BUTLER FOR THE US MILITARY, 1940–1941. ©1960, ESTATE OF BUCKMINSTER FULLER

(Figure 10.5). War-driven developments in materials and production methods made the project more feasible than the 1927 model, and a demobilized bomber factory in Wichita, Kansas, was authorized to produce prototypes for the Air Force. Despite media and customer interest, however, as with the earlier house, a combination of problems shut down the project. Union difficulties, lack of trained installation crews and a viable distribution infrastructure, Fuller's own reluctance to market what he saw only as a prototype and, most fundamental of all, absence of funds for tooling costs, revealed that the strength of design did not guarantee commercial success. A market-driven social revolution looked unlikely.

After the collapse of the Wichita House Fuller moved away from commercial projects into research and education, working as a visiting professor at colleges throughout the USA. Much of his research, in fact, was made possible through the exploitation of facilities and student labour that teaching made available. It is this campus-based work which led to the development of Fuller's most famous innovation, the geodesic dome, a potentially infinitely extendible tensile frame of equilateral triangles. The spherical shape distributes load in all directions throughout the structure and the dome increases in strength the larger it gets.

As an educationalist Fuller found an ideal site for pure research and for spreading the word about an efficient, sustainable, dwelling-led revolution. Certainly, the students were enthusiastic. The communal environment fostered by Fuller's university research projects, so inspirational to the countercultural generation of the 1960s, has, however, overshadowed the fact that Fuller always claimed ownership of work done by his students and that, more seriously, many of the experiments and prototypes were directly utilized by the military. Indeed, as Alex Soojung-Kim Pang has shown, the rise of the geodesic dome, and of Fuller, are largely due to their extensive role in Cold War security and propaganda programmes.

The prospect of nuclear war seemed to threaten Fuller's chances of leading a consumer-sponsored overthrow of the status quo through housing design, since the arms race and

permanent preparedness swallowed up the resources and manpower he had counted on convincing American government and business was needed to help power the non-plan nation. Ironically, however, military strategies being developed to disperse population and industry as preventative and post-apocalyptic measures share much with Fuller's view of unrestricted spatial movement.[12] Supporters of 'dispersal' in government and among architects and town planners also saw the strategy as a way of eliminating poverty, crime and overcrowding in cities and as a means of reinforcing the values of American small-town life. Fuller had imagined for twenty years that his Dymaxion projects would fulfil these very aims. As Kim Pang persuasively suggests, Fuller's popularity rises so radically after two decades of indifference precisely because his ideas are in line with Cold War strategists.[13]

The military had shown interest in the dome as early as 1949 and, by 1953, Fuller was working directly with the Marine Corps and the Air Force. He drew upon student experiments at North Carolina State, Tulane in New Orleans, MIT and the University of Michigan, using materials often donated by industry to produce various hangar and barrack domes. By 1957 the Marines were using some 300 geodesic domes world-wide, including the 55-foot diameter fibreglass radomes stretched along the Distant Early Warning (DEW) line along the northern rim of Canada.

As propaganda, the domes became an essential element of the USA's ideological war during the 1950s, used to house government-sponsored International Trade Fairs in the Third World and behind the Iron Curtain. Consisting of ostentatious displays of American technological prowess and material abundance, the Fuller domes in many places became the central attraction. The success of the large (>100 foot in diameter) trade domes enormously boosted Fuller's reputation at home, where he and his designs were touted in the media as the acme of American individualism and ingenuity.

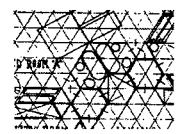
The adoption of Fuller, or Bucky, as he became known, by an anti-institutional counterculture is thus made possible through the visibility his ideas enjoyed as part of the American military-industrial complex's aggressive cultural imperialism during the early years of the Cold War. Although it is possible and perhaps necessary for insurgent forces to refigure the materials and symbols of the status quo, implicit in Fuller's ideas is a form of social engineering which outstrips the smaller freedoms his work more obviously offers. The mobile dwelling machines from the beginning were designed for factory production – Ford was always a hero to Fuller, and Ford's authoritarianism is legend. Only under factory conditions could the necessary precision be guaranteed, as many commune dwellers discovered as they sat beneath leaky handmade versions. The open, undifferentiated, notionally vacant space of the Air-Ocean world might liberate human beings from the stasis of the quotidian, but it also shirks off the bonds of community, of a grounded, accumulated experience of everyday life. When being *here*

12 A.S. Kim Pang, Dome days: Buckminster Fuller in the Cold War, in Francis Spufford and Jenny Uglow (editors), *Cultural Babbage: technology, time and invention,* London: Faber, 1996, pp. 167–192, 172.
13 *Ibid.,* p. 177.

converts into anywhere, identity might fold into itself, having no place, and give itself over to a manufactured vagrant force driven, still, by the unplanned but all-pervasive machinery of capitalism.

Fuller's holistic understanding of global networks of trade and communications, and of the ecological problems written into them, is still pertinent, and still seemingly irresolvable. His response to need assumes an elimination of status and power by a kind of value-free technology before the project can begin. We still live in dead men's houses, if we can find shelter at all. Fuller was right to focus on property as the seat of power, but design science cannot in itself disperse that power, it can merely streamline our confinement.

The House of the Seven Gables concludes with Holgrave's inheritance of a not inconsiderable fortune. Moving to the late Judge Pyncheon's 'elegant country-seat', Holgrave wonders why the judge 'should not have felt the propriety of embodying so excellent a piece of domestic architecture in stone, rather than wood', so that each generation could alter the interior to suit their needs while the exterior 'through the lapse of years, might have been adding venerableness to its original beauty, and thus giving that impression of permanence, which I consider essential to the happiness of any one moment'. Phoebe is amazed at this about-face. With a 'half-melancholy laugh' Holgrave admits: 'You find me a conservative already! Little did I think ever to become one'.[14]



14 Nathaniel Hawthorne, *op. cit.*, note 1, p. 315.